

EFGP log-marginal gradient: a step-by-step derivation

This note derives the EFGP gradient algorithm from scratch. The target form, which we call the β -form, avoids a $1/\sigma^2$ amplification of the CG residual that appears in the naive Woodbury expansion. The derivation is linear: each section uses only what was established in the previous one.

1 Step 1. The log-marginal and its gradient

Let $K_\theta \in \mathbb{R}^{N \times N}$ be the kernel Gram matrix, σ^2 the observation noise, and $\tilde{K} := K_\theta + \sigma^2 I$. The Gaussian log-marginal is

$$\log p(y \mid \theta) = -\frac{1}{2} y^\top \tilde{K}^{-1} y - \frac{1}{2} \log \det \tilde{K} + \text{const.}$$

For any scalar hyper θ (a kernel parameter such as lengthscale or variance, or the noise σ^2 itself), the two standard identities $\partial_\theta \log \det \tilde{K} = \text{tr}(\tilde{K}^{-1} \partial_\theta \tilde{K})$ and $\partial_\theta \tilde{K}^{-1} = -\tilde{K}^{-1} (\partial_\theta \tilde{K}) \tilde{K}^{-1}$ give

$$\partial_\theta \log p = \frac{1}{2} (T_2 - T_1), \quad T_2 = y^\top \tilde{K}^{-1} (\partial_\theta \tilde{K}) \tilde{K}^{-1} y, \quad T_1 = \text{tr}(\tilde{K}^{-1} \partial_\theta \tilde{K}). \quad (1.1)$$

Writing $\alpha := \tilde{K}^{-1} y$,

$$T_2 = \alpha^\top (\partial_\theta \tilde{K}) \alpha. \quad (1.2)$$

So we need two primitives: a single solve $\alpha = \tilde{K}^{-1} y$ (for T_2), and a trace of $\tilde{K}^{-1} \partial_\theta \tilde{K}$ (for T_1). For kernel hypers $\partial_\theta \tilde{K} = \partial_\theta K$; for noise $\partial_{\sigma^2} \tilde{K} = I$.

The obstacle. Both T_1 and T_2 contain $\tilde{K}^{-1}$, an $N \times N$ dense inverse. A direct solve is $\mathcal{O}(N^3)$; the trace is worse. To make these tractable we need structure in K .

2 Step 2. The EFGP factorization

Fix an equispaced frequency grid $\{\xi_k\}_{k=1}^M \subset \mathbb{R}^d$ with spacing h , chosen to resolve the kernel spectrum to tolerance ε . Define

$$F \in \mathbb{C}^{N \times M}, \quad F_{j,k} = e^{i2\pi x_j \cdot \xi_k}, \quad D = \text{diag}(\sqrt{h^d \hat{k}_\theta(\xi_k)}), \quad \Phi := FD. \quad (2.1)$$

Quadrature on the kernel Bochner integral gives the two approximations we need:

$$K_\theta \approx \Phi \Phi^* = F D^2 F^*, \quad \partial_\theta K_\theta \approx F D'_\theta F^*, \quad D'_\theta := \text{diag}(h^d \partial_\theta \hat{k}_\theta(\xi_k)). \quad (2.2)$$

Both approximations share the *same* F ; only the spectral diagonal changes. This is the structural fact that will let us reduce every N -space expression to an M -space expression.

Two auxiliary $M \times M$ operators appear repeatedly:

$$T := F^* F \quad (\text{multi-level Toeplitz, } \mathcal{O}(M \log M) \text{ matvec}), \quad A := \sigma^2 I + DTD = \sigma^2 I + \Phi^* \Phi. \quad (2.3)$$

We will apply CG only against A , never against the $N \times N$ $\tilde{K}$.

3 Step 3. Solving the mean: Woodbury moves the inverse to M -space

Because $\tilde{K} = \sigma^2 I + \Phi \Phi^*$ is a rank- M update of a diagonal, the Sherman–Morrison–Woodbury identity applies directly:

$$\tilde{K}^{-1} = \sigma^{-2} (I - \Phi (\sigma^2 I + \Phi^* \Phi)^{-1} \Phi^*) = \sigma^{-2} (I - \Phi A^{-1} \Phi^*). \quad (3.1)$$

Applying both sides to y ,

$$\alpha = \tilde{K}^{-1} y = \sigma^{-2} (y - \Phi \beta_0) = \sigma^{-2} (y - F D \beta_0), \quad \beta_0 := A^{-1} D F^* y. \quad (3.2)$$

This reduces the mean to one NUFFT ($z := F^* y$) followed by one M -space CG solve ($A \beta_0 = D z$).

A useful reorganization. The equation defining β_0 , $A \beta_0 = D F^* y$, expands (using $A = \sigma^2 I + D T D$) as $\sigma^2 \beta_0 + D T D \beta_0 = D F^* y$. Rearrange:

$$D (F^* y - T D \beta_0) = \sigma^2 \beta_0. \quad (3.3)$$

Separately, applying F^* to (3.2) and using $T = F^* F$ (eq. (2.3)):

$$F^* \alpha = \sigma^{-2} (F^* y - T D \beta_0). \quad (3.4)$$

Multiplying (3.4) by D and comparing with (3.3): $\sigma^2 D F^* \alpha = D (F^* y - T D \beta_0) = \sigma^2 \beta_0$, so

$$\boxed{D F^* \alpha = \beta_0} \quad (\text{exactly, at CG convergence}). \quad (3.5)$$

Equivalently, $F^* \alpha = \beta_0 / D$ entrywise on $\text{supp}(D)$. This identity is the reason the β -form exists: the quantity $F^* \alpha$, which would otherwise require a σ^{-2} -amplified combination of $F^* y$ and $T(D \beta_0)$, is already sitting in memory as β_0 — up to a pointwise division by D .

4 Step 4. T_2 in M -space

Kernel hypers

Substitute (2.2) into (1.2):

$$T_2(\theta) = \alpha^* (F D'_\theta F^*) \alpha = \langle F^* \alpha, D'_\theta F^* \alpha \rangle. \quad (4.1)$$

We now have two algebraically equivalent routes.

Route A (naive Woodbury / residual form). Use eq. (3.4) literally, $F^* \alpha = \sigma^{-2} (F^* y - T D \beta_0)$. Every evaluation carries a σ^{-2} prefactor. Under an inexact CG solve with residual $r := A \beta_0 - D F^* y$, the quantity evaluated is $F^* \alpha + r / (\sigma^2 D)$ (on the D -support). Because (4.1) is *quadratic* in $F^* \alpha$, the error contains a $\|r\|^2 / \sigma^4$ term that can flip the sign of the gradient when σ^2 is small.

Route B (the β -form). Use the identity (3.5): $F^* \alpha = \beta_0 / D$. Then (4.1) becomes

$$\boxed{T_2(\theta) = \langle \beta_0, (D'_\theta / |D|^2) \beta_0 \rangle.} \quad (4.2)$$

This uses only the CG output β_0 . Under inexact CG the error in β_0 enters linearly (through a single factor of β_0 on each side of the inner product), with no $1/\sigma$ prefactor. The $1/|D|^2$ weight is bounded on the EFGP grid by construction (the grid is truncated where $\hat{k}(\xi)$ falls below ε).

Noise

Since $\partial_{\sigma^2} \tilde{K} = I$, we have $T_2(\sigma^2) = \|\alpha\|^2$. Expanding the defining relation $\sigma^2 \alpha = y - FD\beta_0$ directly,

$$\boxed{\|\alpha\|^2 = \sigma^{-4}(\|y\|^2 - 2\Re\langle\beta_0, DF^*y\rangle + \langle\beta_0, DTD\beta_0\rangle).} \quad (4.3)$$

Crucially this expansion does *not* invoke $A\beta_0 = DF^*y$, so it is CG-robust: all three scalars are M -dots on vectors already computed during the mean solve ($F^*y, D\beta_0, T(D\beta_0)$).

Variance σ_f^2 (closed form)

For SE/Matérn kernels, $\hat{k}$ is linear in σ_f^2 , so $D^2 = \sigma_f^2 \cdot (\text{shape})$ and $D'_{\sigma_f^2} = D^2/\sigma_f^2$. Substituting into (4.2):

$$T_2(\sigma_f^2) = \|\beta_0\|^2/\sigma_f^2.$$

5 Step 5. T_1 : trace in M -space

Kernel hypers

By cyclicity of trace and (2.2),

$$T_1(\theta) = \text{tr}(\tilde{K}^{-1}FD'_\theta F^*) = \text{tr}(F^*\tilde{K}^{-1}F \cdot D'_\theta) = \text{tr}(B D'_\theta), \quad B := F^*\tilde{K}^{-1}F \in \mathbb{C}^{M \times M}. \quad (5.1)$$

The trace has moved into M -space. Now apply Woodbury (3.1) to expand B :

$$B = \sigma^{-2}(F^*F - F^*\Phi A^{-1}\Phi^*F) = \sigma^{-2}(T - TDA^{-1}DT). \quad (5.2)$$

Hutchinson. For a Rademacher probe $v \in \{\pm 1\}^M$, $\mathbb{E}_v[v^* B D'_\theta v] = \text{tr}(B D'_\theta) = T_1(\theta)$. Let

$$u^\theta := D'_\theta v, \quad X^\theta := A^{-1}(DTu^\theta).$$

Then (5.2) gives

$$B D'_\theta v = \sigma^{-2}(Tu^\theta - TD X^\theta). \quad (5.3)$$

Route A (naive Woodbury). Evaluate (5.3) literally and inner-product with v . The σ^{-2} prefactor amplifies the CG residual of X^θ .

Route B (the β -form). Apply the same reorganization as in Step 3 to the defining equation $AX^\theta = DTu^\theta$:

$$\sigma^2 X^\theta + DTD X^\theta = DTu^\theta \implies D(Tu^\theta - TD X^\theta) = \sigma^2 X^\theta. \quad (5.4)$$

Multiplying (5.3) by $\sigma^2 D$ and comparing with (5.4): $\sigma^2 D B D'_\theta v = D(Tu^\theta - TD X^\theta) = \sigma^2 X^\theta$, so

$$\boxed{B D'_\theta v = X^\theta/D} \quad (\text{exactly, at CG convergence}). \quad (5.5)$$

Substituting (5.5) into the Hutchinson estimator,

$$T_1(\theta) \approx \frac{1}{J} \sum_{j=1}^J \langle v_j, B D'_\theta v_j \rangle = \frac{1}{J} \sum_{j=1}^J \langle v_j, X_j^\theta/D \rangle. \quad (5.6)$$

One batched CG solve against A produces the $\{X_j^\theta\}$; the rest is a Hadamard divide and an inner product. No NUFFT and no Toeplitz application enter (5.6) after the right-hand side is formed.

Why this is lower-variance than a naive N -space Hutchinson. A data-space Hutchinson estimator $z^* \tilde{K}^{-1} (\partial_\theta K) z$ with $z \in \{\pm 1\}^N$ has its Hutchinson variance proportional to the squared Frobenius norm of the off-diagonal of $\tilde{K}^{-1} \partial_\theta K$. The same trace in M -space is $v^* B D'_\theta v$ with $v \in \{\pm 1\}^M$, and $B D'_\theta$ is $M \times M$. Since $\partial_\theta K \approx F D'_\theta F^*$ has rank at most M by construction, the $N-M$ directions where the N -space matrix acts trivially still contribute Hutchinson variance. The M -space probe sees only the informative subspace: single-probe variance drops by roughly the rank ratio M/N .

Noise

From Woodbury (3.1) applied to the trace,

$$T_1(\sigma^2) = \text{tr}(\tilde{K}^{-1}) = \sigma^{-2} (N - \text{tr}(A^{-1} D T D)), \quad (5.7)$$

with the M -space trace estimated by Hutchinson on $A^{-1} D T D$ using probes $V_n \in \{\pm 1\}^{M \times J}$ solved in the same batched CG.

Variance σ_f^2 (closed form)

Since $K_\theta = \sigma_f^2 \cdot \partial_{\sigma_f^2} K_\theta$,

$$T_1(\sigma_f^2) = \frac{1}{\sigma_f^2} \text{tr}(\tilde{K}^{-1} K) = \frac{N - \sigma^2 T_1(\sigma^2)}{\sigma_f^2}.$$

No separate Hutchinson call.

6 Step 6. Why the β -form everywhere

Both T_2 and T_1 land on the same structural choice: to express the needed quantity ($F^* \alpha$ or $B D'_\theta v$), either

- (A) evaluate the Woodbury expression literally, which carries a σ^{-2} prefactor on the CG residual; or
- (B) apply the reorganization $D \cdot (\text{Woodbury expression}) = \sigma^2 \cdot (\text{CG output})$ and divide by D .

The two are algebraically identical at CG convergence. Under inexact CG with residual δ , Route A contributes $\|\delta\|/\sigma^2$; in T_2 , where this sits on *both sides* of a quadratic form, it becomes $\|\delta\|^2/\sigma^4$ and can flip the gradient sign at small σ^2 . Route B replaces the $1/\sigma^2$ amplification with a $1/D$ division that is bounded on the EFGP grid (the grid is truncated where $\hat{k}(\xi) < \varepsilon$, so $|D|^2 \gtrsim \varepsilon$, and the amplification is $\mathcal{O}(1/\varepsilon)$ rather than $\mathcal{O}(1/\sigma^2)$). This is the entire motivation for the β -form.

7 Step 7. Algorithm

Let $\theta_1, \dots, \theta_{K-1}$ be the kernel hypers excluding σ_f^2 . Precompute F , D , $\{D'_{\theta_i}\}$, and a Jacobi preconditioner for A .

Complexity. One NUFFT ($\mathcal{O}(N + M \log M)$). Two CG calls against A , each $\mathcal{O}(M \log M)$ per matvec: the mean with one RHS, the trace with $(K-1)+1 = K$ blocks of J RHSs stacked. All T_2 work is $\mathcal{O}(KM)$. No NUFFT and no N -space linear algebra appear inside the trace loop.

Algorithm 1 EFGP log-marginal gradient: one NUFFT, two CG calls.

- 1: $z \leftarrow F^* y$ $\triangleright$ **only NUFFT in the whole gradient**
 - 2: Solve $A \beta_0 = Dz$ by CG. **(CG #1, mean)**
 - 3: $w \leftarrow T(D\beta_0)$ $\triangleright$ Toeplitz matvec, reused
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- 4: *Term T_2 (all M -space, Route B where relevant):*
 - 5: kernel hypers: $T_2(\theta_i) \leftarrow \langle \beta_0, (D'_{\theta_i}/|D|^2) \beta_0 \rangle$ [eq. (4.2)]
 - 6: variance: $T_2(\sigma_f^2) \leftarrow \|\beta_0\|^2/\sigma_f^2$
 - 7: noise: $T_2(\sigma^2) \leftarrow \sigma^{-4}(\|y\|^2 - 2 \Re \langle \beta_0, Dz \rangle + \langle \beta_0, Dw \rangle)$ [eq. (4.3)]
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- 8: *Term T_1 :* draw $V \in \{\pm 1\}^{M \times J}$, $V_n \in \{\pm 1\}^{M \times J}$.
 - 9: kernel: for each θ_i , $\text{rhs}^i \leftarrow D T(D'_{\theta_i} V)$
 - 10: noise: $\text{rhs}^n \leftarrow D T(DV_n)$
 - 11: stack RHSs; solve $AX = \text{rhs}$ by batched CG. **(CG #2, trace)**
 - 12: kernel (β -form): $T_1(\theta_i) \leftarrow \text{mean}_j \langle V_j, X_j^i/D \rangle$ [eq. (5.6)]
 - 13: noise: $T_1(\sigma^2) \leftarrow \sigma^{-2}(N - \text{mean}_j \langle (V_n)_j, X_j^n \rangle)$ [eq. (5.7)]
 - 14: variance: $T_1(\sigma_f^2) \leftarrow (N - \sigma^2 T_1(\sigma^2))/\sigma_f^2$
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- 15: **Return** $\nabla_\theta \log p = \frac{1}{2}(T_2 - T_1)$ for each hyper.
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